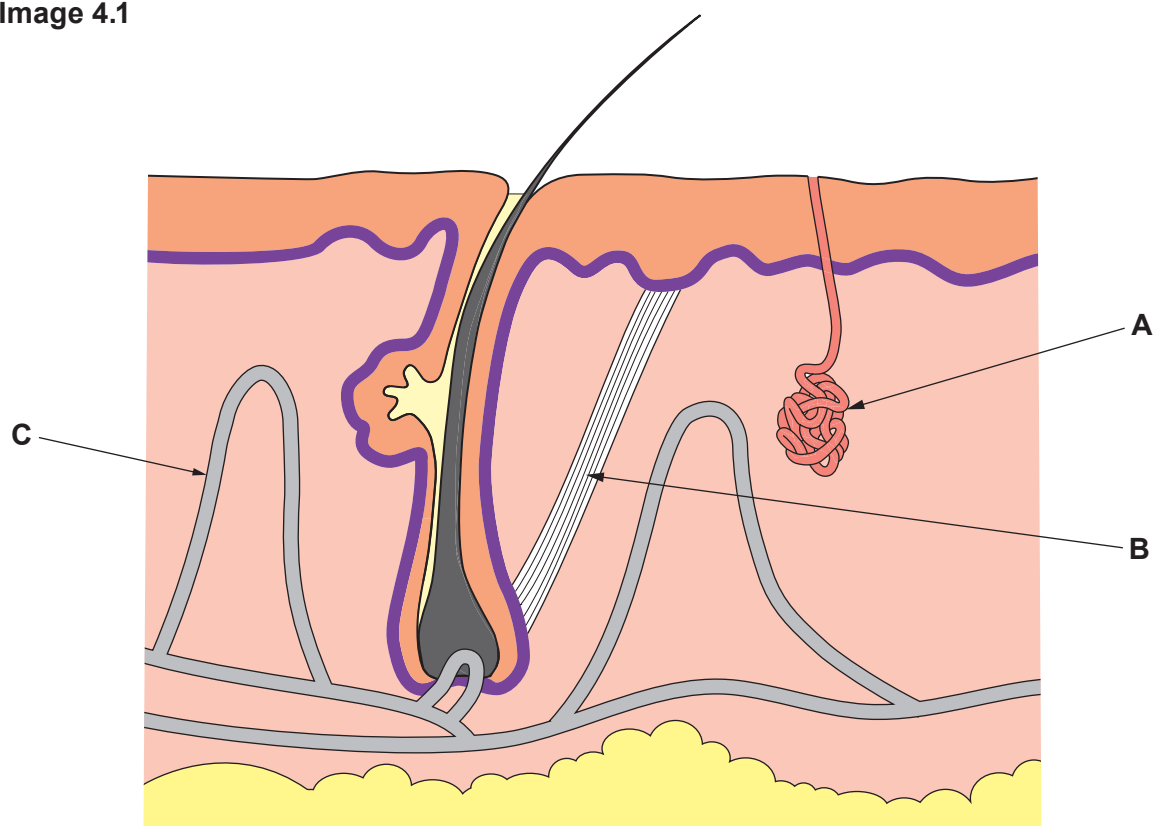


Examiner
only

4. **Image 4.1** shows a section through the human skin.

Image 4.1



- (a) State the name of structures **A** and **B** in **Image 4.1**.

[1]

A

B

- (b) Explain how structure **B** helps to maintain the normal body temperature in cold conditions.

[3]



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- (c) Frostbite occurs when the skin and tissues are damaged by long-term exposure to freezing temperatures. The most common parts of the body that frostbite affects are the fingers, toes, nose, ears, cheeks and chin.
Using **Image 4.1** suggest how the response of the blood vessels labelled **C** to freezing temperatures can lead to frostbite developing in the fingers. [2]

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